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System boundary

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System: Inspection Report Routing System

Inside the boundary

The Inspection Report Routing System does four things: 1. Receive Report accepts the Field Team's inspection report and writes a submitted report to D1 Report Register; 2. Check Required Fields , sends a validation note to the Field Team and passes accepted reports. A rejected report does not continue; 3. Resolve Destination reads routing entries from D2 Routing Table; 4 Send Work Item sends a work item to the resolved department and writes a delivery record to D1.

Outside the boundary

External teams are the Field Team, Electrical, Drainage and Structures departments. Outside the boundary are flight control, physical inspections and repair work. A service administrator maintains the Routing Table.

Boundary flows

One row per information flow. Repeat an entity if it has more than one flow.

External entity	Direction (into / out of system)	Data flow name	Source
Field Team	Into system	Inspection report	W06-S01
Field Team	Out of system	Validation note	W06-S01
Electrical	Out of system	Work item	W06-S01
Drainage	Out of system	Work item	W06-S01
Structures	Out of system	Work item	W06-S01

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Information movement

Processes: 1 Receive Report; 2 Check Required Fields; 3 Resolve Destination; 4 Send Work Item. Stores: D1 Report Register; D2 Routing Table. Use the flow rows to connect these and the external teams from page 1. Eleven flow rows are provided.

From (entity / process / store)	To (entity / process / store)	Data flow name	Source
Field Team	1 Receive Report	Inspection report	W06-S01
1 Receive Report	D1 Report Register	Submitted report	W06-S01
1 Receive Report	2 Check Required Fields	Received report	W06-S01
2 Check Required Fields	Field Team	Validation note	W06-S01
2 Check Required Fields	3 Resolve Destination	Accepted report	W06-S01
D2 Routing Table	3 Resolve Destination	Routing entries	W06-S01
3 Resolve Destination	4 Send Work Item	Report with resolved destination	W06-S01
4 Send Work Item	Electrical	Work item	W06-S01
4 Send Work Item	Drainage	Work item	W06-S01
4 Send Work Item	Structures	Work item	W06-S01
4 Send Work Item	D1 Report Register	Delivery record	W06-S01

Balance check:

Field Team to Receive Report , Check Required Fields to Field Team, and 4 Send Work Item to Electrical, Drainage and Structures. These match the five boundary flows on page 1 in entity, direction and data name. No external team exchanges data directly with D1 or D2. (W06-S01)

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Data definitions and finding

Field	Meaning	Example	Allowed value / format	Source
report_id	Unique reference that identifies one inspection report.	CJ-601	Unique; CJ- followed by three digits.	W06-S02
asset_id	Reference for the inspected asset.	SL-101	Two-letter prefix followed by three digits.	W06-S02
asset_type	The kind of asset that was inspected; the policy uses it to decide the receiving department.	BRIDGE	One of STREETLIGHT, DRAIN or BRIDGE.	W06-S02
finding_code	Code for what the inspection found.	SURFACE_DAMAGE	One of LAMP_OUT, BLOCKED or SURFACE_DAMAGE.	W06-S02
priority	How urgent the report is; it sets urgency within the receiving department and does not change the department.	URGENT	One of ROUTINE or URGENT.	W06-S02
captured_at	The UTC date and time the inspection data was captured.	2026-10-14T09:10:00Z	UTC time written as YYYY-MM-DDTHH:MM:SSZ.	W06-S02
photo_ref	Fictional reference to the photo attachment for the report (no image files are supplied).	IMG-603	IMG- followed by three digits.	W06-S02
survey_area	The fictional survey area the report belongs to (not coordinates).	CENTRAL	One of NORTH, CENTRAL or SOUTH.	W06-S02

Supported finding

CJ-603 is a BRIDGE report (asset_type in W06-S02, row CJ-603). The destination policy says a BRIDGE report belongs to Structures (W06-S03), but the delivery log shows CJ-603 was ACCEPTED and received by Electrical (W06-S04). The expected department (Structures) and the observed department (Electrical) differ, so acceptance by the field check did not mean correct routing.

Unanswered question:

What did the D2 Routing Table entry for BRIDGE contain when CJ-603 was routed? A runtime snapshot of the table or its deployment history would help; W06-S03 states that no such evidence is supplied, so the cause is not established.